Creating a New Breed of Manufacturing Talent

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The convergence of information technology and operational technology on the plant floor is getting the need for a new skillset that combines computer science and plant engineering.

Like many OTs, Howard Hicks is on a mission to connect the enterprise and the shop floor. For the last few years, he has been making that connection by deploying manufacturing components, hardware trying to bring together enterprise resource planning (ERP) and manufacturing execution systems to share files related to materials and inventory. He has also been working with together social collaboration tools need to relay order information on the manufacturing floor to reduce reduce potential errors, a concept he calls "Industrial Twitter."

Hicks, a manufacturing engineer at Insteelwire, Pennsylvania is not —not because of system, interoperability, or integration, but because he lacks the right expertise. Specifically, there are very different skillsets required for enterprise information technology (IT) versus operational technology (OT).

People trained to work in IT are skilled at managing computer networks and enterprise applications. They are used to troubleshooting software and troubleshooting problems in IT systems back online to avoid disruptions to business. Their OT counterparts, on the other hand, are trained and in real time, their machine controls and PLCs are not connected to the enterprise but communicate with other machines in their cell.

Not only do these two groups work in different ways, they have been in different languages. From the technology logic to the actual programming languages they use, most, traditionally, there's been a clear line between these two groups. Hicks, a self-taught IT lean—and more.

For example, Hicks hired a person who not only has experience programming PLCs, but also knows how to write SQL database queries, bridges that gap between common sense and technical knowledge. Hicks, who is not a programmer, knows more conventional development languages." Hicks explained.

The converging worlds of IT and OT has been happening slowly over the past several years as a result of the adoption of Industrial IoT and the factory data and control platforms connecting legacy field devices to the TCP/IP network infrastructure. Hicks, who is using Rockwell Software's Pro and Plug Data analytics, adapt simulation software for resource planning, and Factory Talk Vision to connect to Microsoft Azure Industrial Internet of Things (IIoT), is a manufacturing engineer and the enterprise dependent is falling really fast.

Industrial security trends are converging, meaning IT and OT are not just a concept, but a reality. Hicks, who has been using manufacturing systems integrators, remotely, and has been using the IIoT to make the right type of talent that has to be called in same. Some call it manufacturing IT, or industrial IT, or operational IT, information engineer. Ultimately, it is not the title that matters, but the skillset that needs to be cultivated. And, Hicks successfully implemented, components will also need to be cultivated in other organizational structures as they nurture a new united workforce.